CM1001/CM2007/ CM2011

Computer Lab 1

Agenda

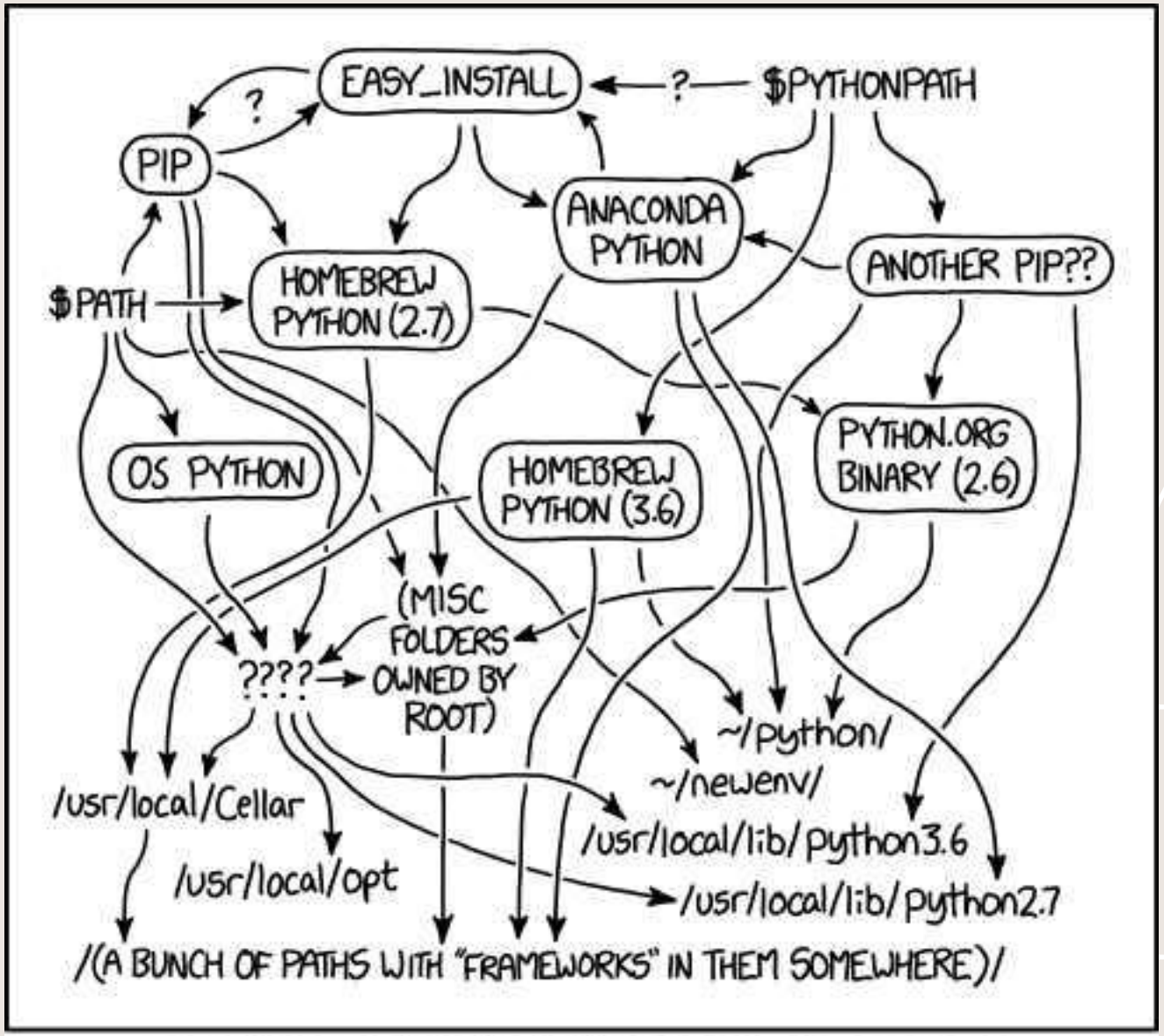
- 10:15-11:00
  - Setting up python and virtual environments
- 11.00-11.15
  - Break
- 11:15-12:00
  - Introduction to common ML libraries (lab1.ipynb)
  - Optional task (see end of lab1.ipynb)
  - Assignment submission structure (last 10 minutes)

Working with python environments

- Don’t mess with system Python.
- Get comfortable with the terminal to manage your Python environments.
- In this course you will be working with .ipynb files. This is a shareable document that combines computer code, plain text (markdown), data, visualizations, graphs and figures, and interactive controls.
- Recommended options for environment management:
  - **Conda (easiest!)**
  - Python built venv (faster)
  - Pyenv + Virtualenv (or virtualenvwrapper)
- IDE:
  - Visual Studio Code (VSC)
  - Jupyter lab or jupyter notebook (this is an extension you add to VSC to work with your .ipynb files)
  - Google Colab (linked to your Google Account)

Working with python environments

- Risk of not starting with proper environment management: imminent & unavoidable factory reset



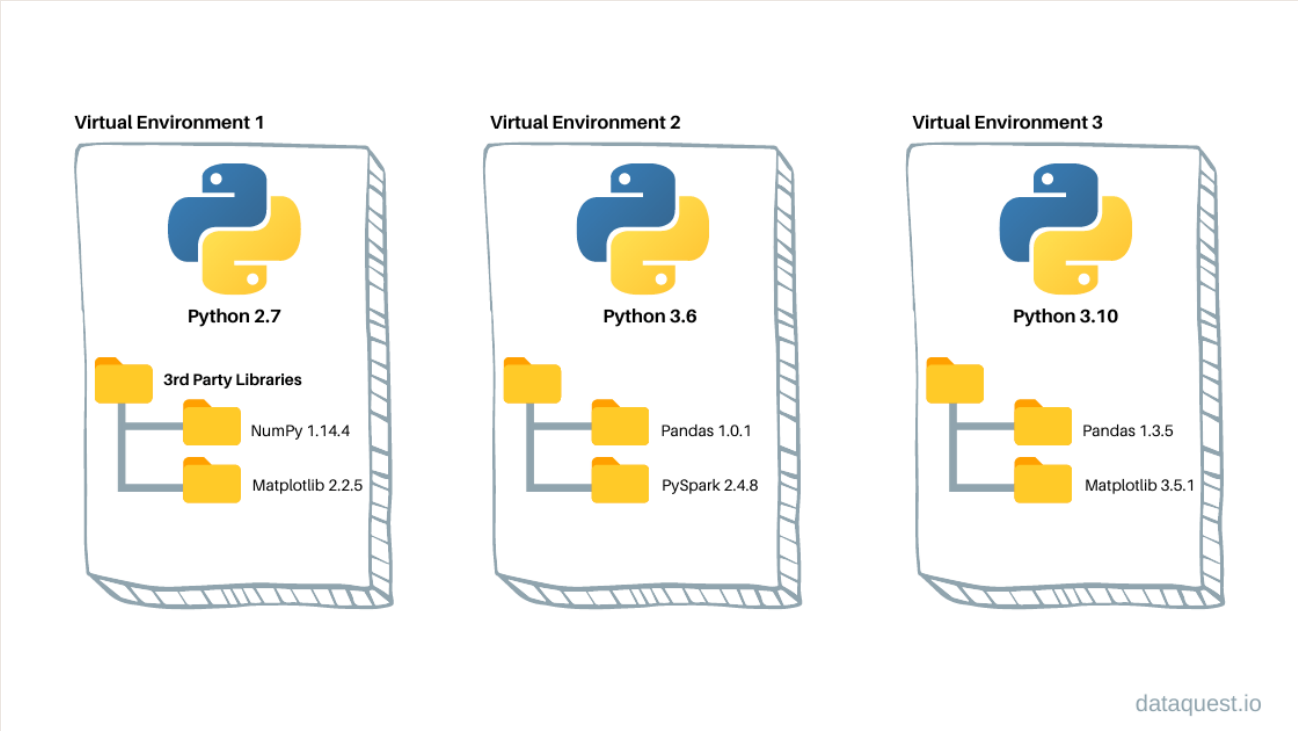
MY PYTHON ENVIRONMENT HAS BECOME SO DEGRADED THAT MY LAPTOP HAS BEEN DECLARED A SUPERFUND SITE.

Source: activestate

Working with python environments

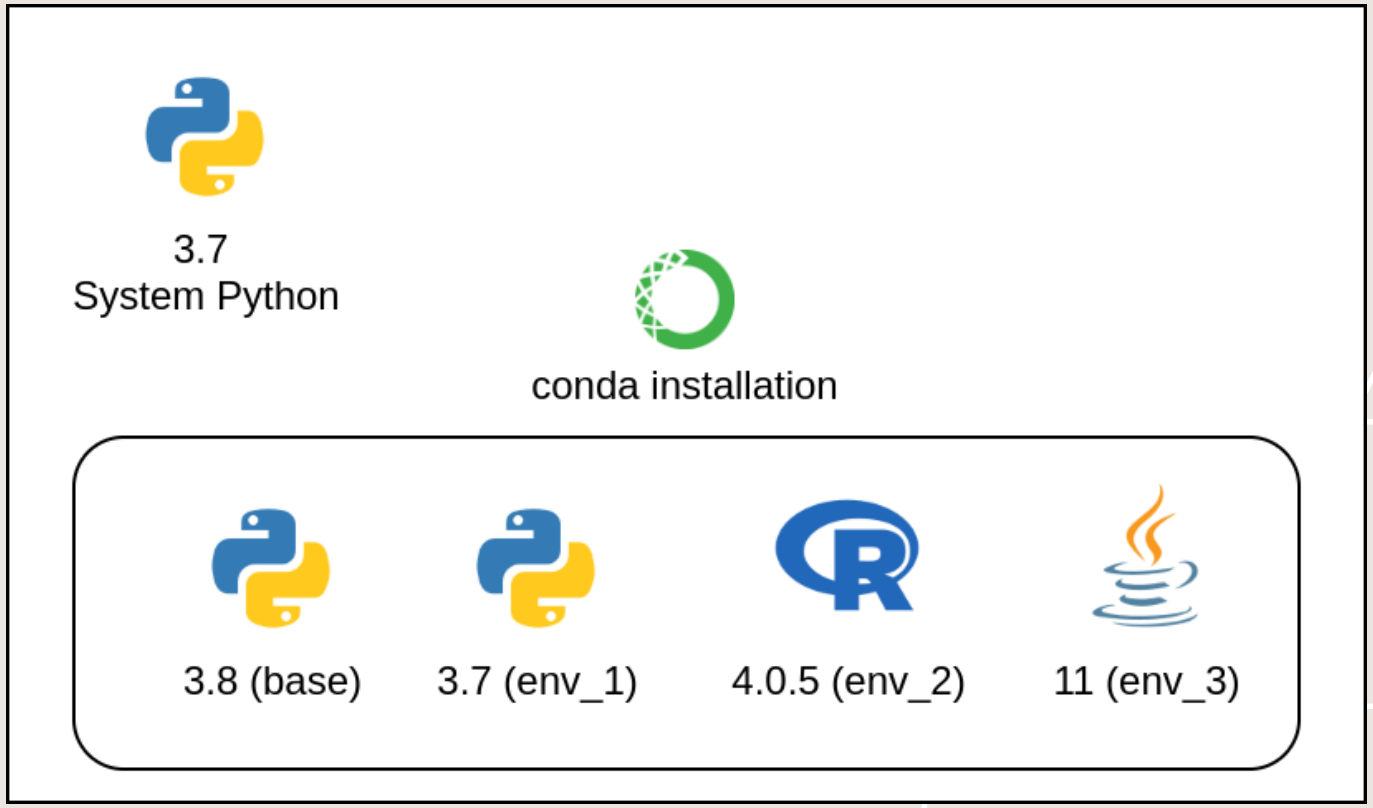
“A Python virtual environment consists of two essential components: the Python interpreter that the virtual environment runs on and a folder containing third-party libraries installed in the virtual environment. These virtual environments are isolated from the other virtual environments, which means any changes on dependencies installed in a virtual environment don’t affect the dependencies of the other virtual environments or the system-wide libraries. Thus, we can create multiple virtual environments with different Python versions, plus different libraries or the same libraries in different versions.”

Source: dataquest



Working with python environments

- Conda:
  - Pro: Easy to set up and use; allows multiple python versions. Comes with a nice user interface Anaconda. Venv:
  - Pro: Easy to set up and use; built-in;
  - Con: Limited in python version
- Pyenv + Virtualenv (or virtualenvwrapper)
  - Pro: allows multiple python versions;
  - Con: Less easy to install;



Set-up Python and Conda

Getting started with Conda: https://conda.io/projects/conda/en/latest/user-guide/tasks/index.html

1. Install python (version 3.11.9 recommended)
    1. (Ubuntu) sudo apt-get install python3.11
    2. (macOS) brew install python@3.11
    3. (windows) https://www.python.org/downloads/windows/

    (windows) REMEMBER TO CHECK THE BOX THAT SAYS ADD TO PATH VARIABLES!!!
2. Open a new terminal and check installation
    1. python –version
3. Install Conda or Anaconda
    1. https://docs.conda.io/projects/conda/en/latest/user-guide/install/index.html
4. Test installation
    1. conda list
5. List existing environments
    1. conda env list
6. Create a new environment
    1. conda create --name envname
    2. conda create --name envname python=3.11.9 (specific python version)
7. Activate environment
    1. conda activate envname
8. Install packages with conda or pip
    1. conda install pkg1 pkg2 ...
    2. conda install pkg=version (specific package version)
    3. conda install --file requirements.txt (packages from a list of requirements)
    4. pip install pkg1 pkg2 ...
    5. pip install pkg=version (specific package version)
    6. pip install -r requirements.txt (packages from a list of requirements)
9. List all installed packages
    1. conda list
    2. pip freeze
10. Create a list of requirements
    1. conda list -e > requirements.txt
    2. pip freeze > requirements.txt
11. Deactivate environment
    1. conda deactivate envname
12. Delete environment
    1. conda remove -n envname –all

Setting up VSC with Jupyter Notebook (first half)

https://www.youtube.com/watch?feature=oembed&v=suAkMeWJ1yE

DYO

Install python and set-up environment for Lab1

Break

See you 11:15

Lab 1 walk-through

Run code together

DYO

Play around with Lab1, ask questions

Assignment submissions

Template and rules